Constraint Geometry and Domain Physics: A Two-Layer Architecture for Binary Partition Diagnostics

Elias De Jesús
Independent Researcher
ORCID: 0009-0007-0190-9143
dejesuselias10@gmail.com

July 2026 — Conceptual Synthesis Note
DOI: 10.5281/zenodo.21695430

Abstract

This note clarifies the mature architecture of a binary-partition research programme without asserting a new theorem or a new physical law. The organizing claim is a division of labor: constraint geometry together with domain physics yields a physical diagnostic. The constraint layer is the conserved binary partition $a + b = 1$ with Thales altitude $h = \sqrt{ab}$, asymmetry $\chi = a - b$ and response $R = 1/(1 - \chi^2) = 1/(4h^2)$, together with intrinsic landmarks such as the crossing $h = a/b$, equivalent on the partition to $a = b^3$. The domain layer supplies the identity of the two sectors, the reason they form a normalized partition, the map $a/b = F(X)$ between a physical variable and the partition coordinates, the trajectory through the manifold, and the empirical meaning of any landmark. The perturbative decomposition of the Einstein tensor into linear and higher-order parts is developed as the primary example: general relativity motivates the schematic sector scaling $a/b \sim C$ with C the compactness, and the intrinsic crossing then reads $C(1 + C)^2 = 1$. The note emphasizes that the geometric landmark is fixed in partition space while its physical coordinate X_* moves with the embedding F, so that an unjustified choice of F can manufacture any desired threshold. A five-step protocol and an explicit epistemic tiering are given, and an illustration in flat ΛCDM separates two failure modes: one landmark there is derived from the dynamics but unoccupied, another occupied but provably mechanism-free.

Keywords: constraint geometry; binary partition; Thales altitude; domain embedding; intrinsic relational landmark; compactness diagnostic; perturbative general relativity; epistemic tiering.

Contents

1 Introduction

1.1 Purpose

A conserved two-sector decomposition appears in many parts of mathematics and physics: a probability allocation, a mass fraction, an energy budget, a density partition, a ratio of operator norms. Whenever two positive quantities are normalized so that they sum to one, the same small piece of geometry becomes available: a bounded state coordinate, a geometric mean, an asymmetry measure, a response measure, and a short list of relational landmarks fixed by the constraint alone.

That availability is not a physical result, and the central purpose of this note is to say precisely what it is instead. The programme summarized here has at various points been described in language that invited a stronger reading than intended — as though partition geometry were generating domain physics. The mature formulation is a division of labor,

$$\boxed{\text{constraint geometry } + \text{ domain physics } \longrightarrow \text{ physical diagnostic}} \tag{1}$$

in which the two layers contribute different and non-interchangeable things. The note is a synthesis and clarification of results developed in earlier work [1, 2, 10, 3, 4, 5]; it introduces no new theorem and no new physical law.

The central interpretive statement is the following.

> *Constraint geometry determines what relational structures and landmarks are available; domain physics determines which of them are physically realized and what they mean.*

1.2 What the two layers supply

The constraint layer supplies the invariant relational structure

$$a + b = 1, \qquad h = \sqrt{ab}, \qquad \chi = a - b, \qquad R = \frac{1}{1 - \chi^2}, \tag{2}$$

together with the coordinate identities among these quantities and any relational condition expressible in them alone.

Domain physics supplies, separately and independently:

(i) the physical meaning of the two sectors;

(ii) the reason those sectors form a genuine normalized partition rather than an imposed one;

(iii) the map between a physical variable and the partition coordinates;

(iv) the trajectory through the partition manifold, if the system evolves;

(v) the empirical meaning, if any, of a geometric landmark.

Neither layer produces a diagnostic alone. Constraint geometry without an embedding is a list of identities; an embedding without independent physical justification is a change of variables chosen to produce a desired answer.

1.3 Non-claims

This note does *not* claim:

(N1) that constraint geometry derives the Einstein field equations, or any other domain field equation;

(N2) that the intrinsic cubic is a universal law of gravity, a fundamental constant of nature, or a universal physical phase transition;

(N3) that nonlinear gravity begins suddenly at the cubic point, or that the point marks a sharp physical onset of any kind;

(N4) that two domains share dynamics because they share a partition;

(N5) that a partition-space landmark has physical meaning prior to an independently justified embedding;

(N6) that the numerical value of any physical threshold obtained here is fixed independently of the schematic normalization of the sector ratio;

(N7) that any of the illustrations in §7 constitutes empirical validation;

(N8) that proximity of an observed value to a landmark is evidence, when the neighbourhood of that landmark contains generic constants within measurement error.

2 Binary Partition Constraint Geometry

Definition 2.1 (Partition data)**.** Let $a+b = 1$ with $a, b > 0$. Define the Thales altitude $h = \sqrt{ab}$, the asymmetry $\chi = a - b \in (-1, 1)$, the response $R = 1/(1 - \chi^2)$, the rapidity $\xi = \operatorname{arctanh} \chi$, and the partition ratio $r = a/b \in (0, \infty)$.

Since $a = \frac{1}{2}(1+\chi)$ and $b = \frac{1}{2}(1-\chi)$, one has $ab = (1-\chi^2)/4$, hence the exact constraint identity

$$4h^2 + \chi^2 = 1, \qquad R = \frac{1}{1-\chi^2} = \frac{1}{4h^2}. \tag{3}$$

Equation (3) is the analytic content of the Thales semicircle: the altitude and the half-asymmetry are the legs of a right triangle on a unit diameter. It is exact, parameter-free, and carries no physics.

Observation 2.2 (Bounded state, unbounded response). The state coordinate is bounded while the response and rapidity are not: $\chi \in (-1,1)$, $R \in [1,\infty)$, $\xi \in \mathbb{R}$. A divergence in R therefore records approach to a partition boundary, in whatever coordinate the domain provides, and is not by itself evidence of a physical singularity [1].

2.1 Intrinsic relational landmarks

A relational landmark is a locus on the partition manifold specified by a condition written in the coordinates (2) alone. The landmark used throughout this note is the altitude–ratio crossing.

Proposition 2.3 (Altitude–ratio crossing; exact). *On $a+b=1$ with $a,b>0$,*

$$h = \frac{a}{b} \quad \Longleftrightarrow \quad a = b^3 \quad \Longleftrightarrow \quad b^3 + b - 1 = 0. \tag{4}$$

The cubic b^3+b-1 is strictly increasing on $[0,1]$ with a sign change, so the crossing occurs at exactly one interior state, and

$$b_* = 0.682327803828019\ldots, \qquad a_* = b_*^3 = 0.317672196171981\ldots, \tag{5}$$

$$h_* = b_*^2 = r_* = \frac{a_*}{b_*} = 0.465571231876768\ldots = \psi - 1, \tag{6}$$

where $\psi = 1/b_* = 1.465571\ldots$ satisfies $\psi^3 = \psi^2 + 1$. In supergolden coordinates $(a_*, h_*, b_*) = (\psi^{-3}, \psi^{-2}, \psi^{-1})$, with $\chi_* = a_* - b_* = -0.3646556\ldots$ and $R_* = \psi^4/4 = 1.1533676\ldots$

Proof. If $h = a/b$ then $ab = a^2/b^2$, so $ab^3 = a^2$ and, dividing by $a > 0$, $b^3 = a$. Conversely $a = b^3$ gives $h = \sqrt{b^3 \cdot b} = b^2$ and $a/b = b^2$. Substituting $a = 1-b$ gives $b^3 + b - 1 = 0$; the derivative $3b^2 + 1 > 0$ gives uniqueness. The identity $\psi - 1 = \psi^{-2}$ follows from $\psi^3 = \psi^2 + 1$ on division by ψ^2; the closed forms for h_* and R_* then follow from (3). These facts are established in [3]. □

Lemma 2.4 (Sign law; exact). *On the partition, $h > a/b \iff b^3 > a \iff b > b_*$, and $h < a/b \iff b < b_*$. The crossing is therefore not a tangency: the two functions cross transversally, in the sense that the difference changes sign there.*

Remark 2.5 (What the landmark is, and is not). Proposition 2.3 is a statement about two functions of one constrained variable. It says that the geometric mean of the two sectors and their ratio agree at exactly one interior state. It does not say that anything happens there. Nothing in (4) refers to a physical system, and the number $0.465571\ldots$ is at this stage an algebraic constant of the partition, on the same footing as the apex value $h = \frac{1}{2}$ at $\chi = 0$.

Remark 2.6 (A caution on levels). The altitude is a geometric mean of the two fractions — an amplitude-level object — while the ratio a/b compares them directly. Setting the two equal is a partition-intrinsic relational condition; it is not a balance law between two physical quantities of like dimension, and should not be read as one. This caution is the reason the crossing is called a *landmark* rather than an equilibrium or a threshold in this section [4].

3 Domain Physics and Physical Embeddings

Definition 3.1 (Domain embedding)**.** Let X be a physical variable of a domain, and let the domain identify two sectors whose normalized fractions are a and b. A *domain embedding* is a map

$$\frac{a}{b} = F(X), \qquad a = \frac{F(X)}{1 + F(X)}, \qquad b = \frac{1}{1 + F(X)}, \tag{7}$$

so that a trajectory in X becomes a trajectory on the partition manifold.

Two things must be supplied from outside the geometry before (7) means anything.

First, the *partition itself* must be genuine. The two sectors must exhaust a conserved or otherwise fixed total for a reason internal to the domain, not because they were divided by their sum for convenience. Any two positive quantities can be normalized; a partition is physically meaningful only when the normalization corresponds to a conservation law, a closure relation, an exact decomposition, or a definitional completeness.

Second, the *functional form of* F must be independently motivated. This is the load-bearing requirement of the whole architecture, and §5 shows why: because the geometric landmark is fixed in partition space, the physical location of that landmark is determined entirely by F. An unconstrained F can place a threshold wherever one wishes.

Observation 3.2 (Division of labor). Constraint geometry fixes the set of available relational landmarks and their locations in partition space. Domain physics fixes which sectors are being partitioned, the embedding F, the trajectory, and the interpretation. A physical diagnostic exists only where both are supplied, and only survives scrutiny when the second is independently justified.

4 The Perturbative General-Relativity Compactness Example

This section develops the motivating example. It is the case in which the partition is most clearly not imposed by hand, because the two sectors come from an exact rearrangement of a field equation.

4.1 An exact two-sector decomposition

Write $g_{\mu\nu} = \bar{g}_{\mu\nu} + h_{\mu\nu}$ with $\bar{g}$ a background and h the perturbation. The Einstein tensor separates exactly into its part linear in h and the remainder,

$$G_{\mu\nu}[g] = \underbrace{G^{(1)}_{\mu\nu}[h]}_{\text{linear}} + \underbrace{G^{(\geq 2)}_{\mu\nu}[h]}_{\text{higher order}}, \tag{8}$$

so that the field equation may be rewritten as

$$G^{(1)}_{\mu\nu}[h] = 8\pi G \left(T_{\mu\nu} + t_{\mu\nu}\right), \qquad t_{\mu\nu} \equiv -\frac{1}{8\pi G}\, G^{(\geq 2)}_{\mu\nu}[h]. \tag{9}$$

Equation (9) is a rearrangement, not an approximation: the matter source and the gravitational self-energy contribution together form the effective source of the linear operator. The object $t_{\mu\nu}$ is gauge-dependent and is related in harmonic gauge to the standard gravitational stress pseudotensor [15, 16].

This is what makes the two-sector reading legitimate here: the domain, not the geometry, says that there are exactly two channels and that they exhaust the effective source.

4.2 The schematic sector scaling

For a quasi-spherical configuration with compactness

$$C = \frac{2GM}{rc^2}, \tag{10}$$

the perturbation scales as $|h| \sim C$, and the leading higher-order terms of the Einstein tensor, being quadratic in h and its derivatives, motivate the schematic estimate

$$\frac{\text{nonlinear sector}}{\text{linear sector}} = \frac{\|G^{(\geq 2)}\|}{\|G^{(1)}\|} \sim C. \tag{11}$$

Statement (11) belongs to standard perturbation theory [12, 16] and is not derived here. It is schematic in two respects that matter later: it is an order-of-magnitude relation between norms rather than an equality, and both norms are gauge- and slicing-dependent.

Remark 4.1 (Structural versus orbital compactness). Two distinct quantities share the algebraic form (10) and are routinely both called compactness, and the distinction matters for anything said later about perturbative accuracy. In (10) as used here, r is a radius of the configuration itself, so C is a *structural* quantity: a fixed property of a star or a static geometry. In a compact binary, the post-Newtonian expansion is instead controlled by the orbital parameter $v^2/c^2 \sim GM/(rc^2)$ with r the orbital *separation*, a *dynamical* quantity that sweeps upward through the inspiral. The two are not the same object, and a landmark value of one is not a landmark value of the other. Consequently, an observation that some numerical value is close both to the compactness of massive neutron stars and to the regime where post-Newtonian accuracy degrades is not one coincidence but two claims about two quantities that happen to share a symbol. This note keeps the structural reading throughout, and makes no claim about the orbital one.

4.3 Normalization and the partition

Normalizing the two sectors to a partition, and writing the ratio as the embedding $F(C) = C$,

$$\frac{a}{b} = C, \qquad a = \frac{C}{1+C}, \qquad b = \frac{1}{1+C}, \qquad a + b = 1. \tag{12}$$

This is a domain embedding in the sense of Definition 3.1. Its elementary properties are: $a < b$ for $C < 1$, so the higher-order sector is the minority channel in the sub-horizon range; $a = b$ at $C = 1$; and $a \to 0$ as $C \to 0$, recovering a single-channel linear description in the weak field.

Independently of (12), the constraint geometry of §2 supplies

$$h = \sqrt{ab} = \frac{\sqrt{C}}{1+C}. \tag{13}$$

4.4 The intersection

Imposing the intrinsic crossing condition $h = a/b$ of Proposition 2.3 on the embedded partition gives, in partition coordinates, $a = b^3$, and in the physical coordinate

$$\frac{\sqrt{C}}{1+C} = C \quad \Longrightarrow \quad C(1+C)^2 = 1 \quad \Longleftrightarrow \quad C^3 + 2C^2 + C - 1 = 0, \tag{14}$$

whose unique positive root is

$$C_{\text{crit}} = \psi - 1 = 0.465571231876768\ldots = h_* = r_*. \tag{15}$$

Derivation of (14). From (13), $h = C$ reads $\sqrt{C} = C(1+C)$; squaring gives $C = C^2(1+C)^2$ and dividing by $C > 0$ gives $C(1+C)^2 = 1$. Equivalently, substituting (12) into $b^3 = a$ gives $(1+C)^{-3} = C(1+C)^{-1}$, i.e. the same polynomial after multiplication by $(1+C)^3$. □

Remark 4.2 (The two routes are one route). The coincidence $C_{\text{crit}} = h_*$ in (15) is algebraic, not evidential. Both computations impose the same condition $a = b^3$ on the same partition and differ only in which coordinate is solved for; the compactness polynomial $C^3 + 2C^2 + C - 1$ and the altitude polynomial $h^3 + 2h^2 + h - 1$ are the same polynomial. This equality is therefore not an independent cross-check, and it is recorded here so that it is not mistaken for one [4].

4.5 The division of labor in this example

Ingredient	Supplied by
Two sectors exhausting the effective source	the exact rearrangement (9)
The sector ratio $a/b = C$	general relativity, via (11)
The altitude $h = \sqrt{ab}$	constraint geometry
The intrinsic condition $h = a/b$, i.e. $a = b^3$	constraint geometry
The cubic $C(1+C)^2 = 1$ and its root	the intersection of the two
Any physical meaning of that root	neither, so far

Structural Interpretation 4.3. General relativity supplies the physical sector ratio. Constraint geometry supplies the intrinsic consistency condition. Their intersection supplies a cubic compactness diagnostic — a candidate diagnostic, with a definite number attached, and nothing more until the embedding is independently justified and the diagnostic tested.

Remark 4.4 (Explicit disclaimers for this section). Constraint geometry does not derive the Einstein field equations; the uniqueness of $G_{\mu\nu} + \Lambda g_{\mu\nu}$ is a separate matter with a separate proof [13, 11]. The cubic (14) is not a universal law of gravity. And C_{crit} is not a point at which nonlinear gravity begins: the higher-order sector is nonzero for every $C > 0$ and grows smoothly, so no onset is asserted. What (14) locates is where a particular relational condition on a particular normalized decomposition is met.

As to accuracy: the post-Newtonian expansion is reliable during the weak-field, slow-motion inspiral and progressively loses accuracy as a binary approaches the strong-field merger regime, where resummation methods or numerical-relativity calculations become necessary [14]. That statement is qualitative, concerns the orbital parameter rather than the structural compactness of (10) (Remark 4.1), and attaches no numerical onset to any value. Nothing here asserts that accuracy degrades near C_{crit}, and no such claim should be read into (14): establishing one would require a dedicated study specifying the gauge, the waveform family, the post-Newtonian order, and the error measure against which degradation is defined. Note also that "loses accuracy" is the defensible claim and is weaker than "diverges": the post-Newtonian series is an asymptotic expansion whose convergence properties are not established, so what degrades is truncation accuracy at fixed order.

5 Intrinsic Marker Versus Physical Realization

This section states the conceptual distinction that the rest of the note depends on.

The intrinsic marker is fixed in partition space:

$$a = b^3, \qquad a + b = 1, \qquad \text{hence} \qquad r_* = \frac{a_*}{b_*} = \psi^{-2} = 0.465571\ldots \tag{16}$$

Its physical realization, however, is the solution of

$$F(X_*) = r_*, \tag{17}$$

which depends on the embedding. The geometry does not move. The physical coordinate X_* moves whenever F changes.

Three distinct objects must therefore be kept apart:

(1) The intrinsic geometric marker. The locus $a = b^3$ on the partition, with its fixed values $(a_*, h_*, b_*) = (\psi^{-3}, \psi^{-2}, \psi^{-1})$. Exact, domain-free, and carrying no physical content (Remark 2.5).

(2) The domain-specific embedding. The map $a/b = F(X)$, which converts the marker into a value X_* of a physical variable via (17). This is where all the physics of the identification lives, and where all the freedom to be wrong lives.

(3) The physical interpretation. A claim that X_* names something — a transition, a bound, a degradation of an approximation, an observable clustering. This requires evidence beyond (1) and (2).

Observation 5.1 (The threshold is as firm as the embedding). Suppose the sector ratio is more accurately $a/b = \alpha C$ for some dimensionless α not fixed by (11), which is an order-of-magnitude statement between gauge-dependent norms. Then (17) gives $C_* = r_*/\alpha$, and the physical threshold scales inversely with a quantity the derivation never pinned down. More generally, for $F(C) = \alpha C^n$ one obtains $C_* = (r_*/\alpha)^{1/n}$. The intrinsic marker is exact; the number 0.4656 is exact *as a partition ratio*; the compactness value inherits the full indeterminacy of the normalization of (11).

Safeguard 5.2 (The manufacturing objection). Given any target value X_0 and the fixed r_*, the embedding $F(X) = r_* X/X_0$ places the landmark exactly at X_0. Therefore no agreement between a partition landmark and an observed value is evidence for anything unless F was fixed, in full, by domain physics before the comparison was made. This is the single most important safeguard in the programme, and it is the reason the perturbative-GR case is the primary example: there, $F(C) = C$ follows from an operator estimate that exists independently of any partition consideration — up to precisely the normalization ambiguity of Observation 5.1, which must be reported rather than suppressed.

6 Relation to the Two-Face and Effective-Potential Frameworks

The architecture (1) is a refinement of an earlier intuition, not a replacement for it. The progression can be described in three stages.

Stage 1: the Two-Face Problem [1]. Binary partition geometry was introduced as a translation framework. The same constrained state admits a bounded description in χ, a hyperbolic description in $\xi = \operatorname{arctanh} \chi$, and a radial or response description in $R = 1/(1-\chi^2)$; apparent conflicts between descriptions of a system can be conflicts between faces rather than between physical claims. In the present language, Stage 1 established that the constraint layer possesses a well-defined coordinate vocabulary, and that divergences in one face may be boundary effects rather than physical singularities.

Stage 2: effective potentials on the partition manifold [2]. The Fisher metric of the binary partition,

$$ds_F^2 = \frac{da^2}{a} + \frac{db^2}{b} = \frac{d\chi^2}{1-\chi^2} = R\, d\chi^2, \tag{18}$$

was proposed as a universal kinetic skeleton, with a domain potential $V_{\text{domain}}(\chi)$ carrying the contextual dynamics. The same split appears in the first-integral extraction as

$$V_{\text{eff}}^{(1)}(\chi) = -\tfrac{1}{2}\,\frac{(\partial_x \chi)^2}{1-\chi^2} = -\tfrac{1}{2}\,\frac{\text{contextual numerator}}{\text{universal denominator}}, \tag{19}$$

which is the dynamical instance of the architecture:

$$\begin{gathered}\text{universal Fisher denominator} \;+\; \text{contextual numerator} \\ \longleftrightarrow \\ \text{universal constraint structure} \;+\; \text{domain-specific dynamics.}\end{gathered} \tag{20}$$

Stage 2 also established the discipline that the present note inherits: applied to a single known trajectory, the extraction (19) is tautological, because the free function V has as many degrees of freedom as the data. Single-trajectory extraction is therefore a Tier-1 normal form and acquires falsifiable status only after an over-determination test — a shared-V family test with a reported residual, an independently determined V that then predicts the trajectory, cross-domain recurrence under a pre-declared normalization, or genuine compression [2].

Stage 3: perturbative GR and the intrinsic cubic [4, 3, 5]. The constraint layer was shown to supply not only coordinates and a kinetic skeleton but also intrinsic relational *landmarks* — loci fixed by the constraint alone — with general relativity supplying an embedding under which one such landmark acquires a compactness reading.

Structural Interpretation 6.1 (Refinement, not abandonment). The original intuition was that a single constrained geometry underlies structurally similar relations across domains. That intuition survives, in a sharpened form: what recurs is the *state space*, its coordinate vocabulary, and its landmark structure — not the dynamics, not the mechanisms, and not the physical meanings. The mature formulation is:

> *The partition framework does not generate domain physics. It provides a common constrained state space, a coordinate language, and a set of intrinsic relational landmarks against which domain-specific dynamics can be organized and tested.*

Remark 6.2 (A coordinate caution carried forward). The flat coordinate of the Fisher metric (18) is $u = \arcsin \chi$, not the rapidity ξ, which is flat for the distinct hyperbolic metric $d\chi^2/(1-\chi^2)^2$. Writing a Fisher-denominator kinetic term in ξ therefore leaves a residual $\operatorname{sech}^2 \xi$ weight that is a signature of the metric–coordinate pairing and not of domain physics [2]. Comparisons across domains must declare the pairing.

7 Cross-Domain Illustrations

These illustrations are deliberately brief. Their purpose is to show that the same intrinsic marker occurs in each partition while its physical status differs completely — which is exactly the point of the architecture. The perturbative-GR case of §4 remains the central example.

7.1 Flat ΛCDM: two landmarks, failing in opposite ways

In a flat cosmology the density parameters exhaust unity, so the partition is genuine rather than imposed: the closure is a constraint of the Friedmann system, not a normalization of convenience. In the matter-plus-Λ regime where radiation is negligible, take

$$a = \Omega_m, \qquad b = \Omega_\Lambda, \qquad a + b = 1, \tag{21}$$

so that the partition geometry organizes the coupling $\mu = ab = \Omega_m \Omega_\Lambda$, the asymmetry $\chi = \Omega_m - \Omega_\Lambda$, and the response $R = 1/(1-\chi^2)$.

Remark 7.1 (Which sector is a, and why it matters). The condition $a = b^3$ is not symmetric under exchange of the sectors, whereas $h = \sqrt{ab}$, χ^2 and R are. Identifying a landmark by its response value therefore specifies an unordered *pair* of states: $R = 1.15337\ldots$ is attained both where $\Omega_m = \Omega_\Lambda^3$ and where $\Omega_\Lambda = \Omega_m^3$, and these are different epochs. Only the ordered condition selects a branch, and only domain physics can order the sectors. The assignment (21) is the one under which the occupied state is the marker: matter is at present the minority sector, with $\Omega_m \approx 0.317 \approx a_*$, which is also the orientation of the general-relativity embedding of §4, where the higher-order sector is the minority channel. It additionally aligns the sign convention, since $\chi_* < 0$ in Proposition 2.3 and $\chi_{\text{today}} = \Omega_m - \Omega_\Lambda < 0$. Reversing the assignment moves the marker to a substantially earlier epoch and flips the sign of χ; nothing in the constraint geometry chooses between the two orderings.

The embedding is supplied by the conservation laws rather than chosen. With scale factor s, matter dilution $\rho_m \propto s^{-3}$ and ρ_Λ constant give

$$\frac{a}{b} = F(s) = \frac{\Omega_{m,0}}{\Omega_{\Lambda,0}}\, s^{-3}, \tag{22}$$

and, more generally, the rapidity $\xi_{\text{DE}} = \frac{1}{2}\ln(\rho_{\text{DE}}/\rho_m)$ obeys

$$\frac{d\xi_{\text{DE}}}{d\ln s} = -\tfrac{3}{2}\, w_{\text{DE}}(s), \tag{23}$$

which follows from the two separate conservation laws alone — $w_m = 0$ and $\rho'_{\text{DE}}/\rho_{\text{DE}} = -3(1 + w_{\text{DE}})$ — and requires neither flatness nor two-component closure [6]. This is an embedding of the stronger kind: it is the independent-determination route of [2], with ΛCDM the case $w_{\text{DE}} = -1$, for which the flow is a uniform translation in rapidity at rate $3/2$.

Two landmarks are now available on this trajectory, and they fail in opposite ways.

(a) A landmark with a mechanism and no occupancy. The domain flow supplies its own landmark. Writing $J = \mu|\chi|$, flat ΛCDM satisfies exactly

$$\frac{d\Omega_m}{d\ln s} = -3\mu, \qquad \frac{d^2\Omega_m}{(d\ln s)^2} = 9\mu\chi = 9J\operatorname{sgn}\chi, \tag{24}$$

so the coupling is the velocity of the matter-fraction flow and the coupling–asymmetry product is its acceleration; the acceleration extremum lies at $|\chi| = 1/\sqrt{3}$, i.e. at $R_J = 3/2$, equivalently at the state where $d^3\Omega_m/(d\ln s)^3$ vanishes [6]. This landmark is *derived from the domain mechanism* and therefore means something about the flow. The present epoch is not there: R_J is attained at $z \approx 1.0$ and $z \approx -0.17$, some twenty standard deviations from today in the cosmological parameters [6, 19].

(b) A landmark with occupancy and no mechanism. The intrinsic marker $a = b^3$ is realized at $F(s_*) = r_*$, i.e. at

$$s_* = \left(\frac{\Omega_{m,0}}{r_*\,\Omega_{\Lambda,0}}\right)^{1/3}, \tag{25}$$

which for concordance parameters falls at $z \approx 0.002$ — effectively now, with $R_* = 1.15337$ against a present-day $R \approx 1.156$ [6, 19]. Here occupancy is excellent and the mechanism is absent; and in this domain the absence can be *proved* rather than merely conceded. Using $d\chi/d\ln s = \frac{3}{2}(1-\chi^2)$, the flow admits an exact Fisher-gradient representation

$$\frac{d\chi}{d\ln s} = -(1-\chi^2)\,U'(\chi), \qquad U(\chi) = -\tfrac{3}{2}\chi, \tag{26}$$

in which the potential is linear, so $U' \equiv -\frac{3}{2} \neq 0$ everywhere. The flow therefore has no interior critical point at all: its only fixed points are the simplex edges $\chi = \pm 1$, and at the cubic state $d\chi/d\ln s = \frac{3}{2}(1-\chi_*^2) \approx 1.30 \neq 0$ [6]. The universe passes through the cubic state at ordinary speed. A mechanism that selected it would require a domain potential with an interior minimum there, which is not ΛCDM.

Structural Interpretation 7.2 (The sharpest form of the thesis). Neither cosmological landmark is a diagnostic, and they miss for opposite reasons. Landmark (a) is derived from the domain and unoccupied; landmark (b) is intrinsic to the constraint geometry and occupied but provably mechanism-free. Meaning and occupancy are supplied by different layers, and neither implies the other. A framework that reported only (b) would look like a success; a framework that reported only (a) would look like a mechanism. Reporting both is what makes the architecture of (1) visible.

7.2 The effective-potential framework

Under a pullback $\chi = \chi_{\text{domain}}(x)$ the same split reappears dynamically as (19): the denominator $1-\chi^2$ is contributed by the chart, and the numerator $(\partial_x\chi)^2$ by the domain approach law. Shared boundary divergence as $|\chi| \to 1$ is therefore a shared response geometry and not shared physics; two systems can share the boundary class while occupying different numerator classes [2]. Single-trajectory extraction remains a Tier-1 normal form unless it passes an over-determination test, as recalled in §6.

7.3 Newtonian two-body geometry

Fixed total mass gives a genuine partition. With $c = m_1 + m_2 = M$ and segments m_1, m_2, the altitude theorem gives $h^2 = m_1 m_2$ exactly, so the gravitational force and potential energy are altitude-squared quantities, $F_{\text{grav}} = Gh^2/r^2$ and $U = -Gh^2/r$, while the reduced mass is $\mu = h^2/c$ and the symmetric mass ratio is $\nu = h^2/c^2$ [10]. In normalized form $a = m_1/M$, $b = m_2/M$ the embedding is the mass ratio itself, $F(q) = q$ with $q = m_1/m_2$, so the intrinsic marker is realized at the single mass ratio $q_* = r_*$, i.e. $m_2/m_1 = 1/r_* = \psi^2 = 2.1479\ldots$

This is the most instructive of the three illustrations precisely because nothing happens there. Newtonian mechanics supplies the physical laws and the observables; it assigns no distinguished role to that mass ratio. The intrinsic marker exists in the partition, is exactly located, and is physically empty in this domain. Meanwhile the same geometry does useful organizational work elsewhere in the same system: it exhibits the coupling–timing asymmetry whereby force is altitude-governed and Kepler timing is hypotenuse-governed, which is an exact re-expression of standard relations and not a new result [10].

Observation 7.3 (Landmark availability versus landmark realization). Across the illustrations the marker $a = b^3$ is available in every case and means something different in each: a candidate diagnostic under a schematic operator embedding (§4); an occupied but provably mechanism-free transit state under an independently derived evolution law (§7.1(b)); and nothing at all under a trivially exact embedding (two-body masses). Alongside it, ΛCDM carries a second landmark that is mechanism-bearing and unoccupied (§7.1(a)). Availability is geometric; realization is physical; and the two can be present separately.

8 A Protocol, and its Safeguards

The following is a conservative working protocol. Steps 1–4 are the constructive part; step 5 is what makes the result scientific rather than decorative.

Step 1. Identify a genuine constrained decomposition. Locate two sectors that exhaust a conserved total, an exact operator decomposition, or a definitionally complete budget, for reasons internal to the domain. Reject partitions created by dividing two unrelated quantities by their sum.

Step 2. **Normalize to $a + b = 1$.** Record the normalization explicitly, including any gauge, slicing, or norm choice on which it depends.

Step 3. **Derive the domain map.** Obtain $a/b = F(X)$ from domain physics — a conservation law, an evolution equation, an operator estimate, an equation of state — and state which parts of F are derived and which are conventional. In particular, report any undetermined overall normalization, since by Observation 5.1 it propagates directly into the location of any landmark.

Step 4. **Evaluate the landmarks.** Solve $F(X_*) = r_*$ for the intrinsic marker of interest, and ask whether X_* is a physically meaningful coordinate: does the domain independently distinguish it? Is it inside the physically admissible range? Does it correspond to any observable change?

Step 5. **Require independent justification or falsifiable testing of the embedding.** An arbitrary choice of F can manufacture any desired threshold (Safeguard 5.2). The embedding must therefore either follow from independent domain physics, or the resulting diagnostic must make a prediction that could fail: a held-out prediction, a collapse of a family of trajectories onto one structure with a reported residual, a cross-domain recurrence under a pre-declared normalization, or a compression of a complex description into a shorter one [2].

8.1 Additional safeguards

Safeguard 8.1 (Universality trap). Many systems normalize to $a + b = 1$. Shared arithmetic is not shared physics, and the recurrence of the partition across domains is expected rather than informative.

Safeguard 8.2 (Chart contributions). Divergences and boundary behaviour contributed by the coordinates — $R \to \infty$ as $|\chi| \to 1$, the Fisher denominator, the response face generally — must not be reported as domain findings.

Safeguard 8.3 (Algebraic identities are not cross-checks). When two computations impose the same condition on the same partition and differ only in which coordinate is solved for, their agreement is a tautology (Remark 4.2). Cross-checks must involve independent inputs.

Safeguard 8.4 (Admissible range). A partition coordinate normalized to $[0, 1]$ need not have a physically admissible range that large. Any landmark must be located relative to the domain's own bounds, and a landmark outside them is not a diagnostic.

Safeguard 8.5 (Landmark crowding). Before reporting a landmark as occupied, check whether its neighbourhood contains generic constants within the measurement error of the domain. Near the cubic value the interval of width comparable to current cosmological precision contains at least three unrelated numbers: the cubic root $b_* = 0.682328$; the Gaussian one-sigma mass $\mathrm{erf}(1/\sqrt{2}) = 0.682689$, a generic aggregation landmark with no domain-specific content [9]; and the present-day $\Omega_{\Lambda,0} \approx 0.6835$ with an uncertainty of order 0.007 [19, 6]. The separations are an order of magnitude below the uncertainty, so proximity cannot discriminate among them and agreement with any one of them is not structural. The safeguard is general: a landmark is only as informative as its neighbourhood is sparse.

Safeguard 8.6 (Information-content check). Before testing whether a partition coordinate organizes an observable, check whether it can. If every partition coordinate is a deterministic invertible function of one domain quantity, then all of them carry exactly the information already in that quantity and none adds a predictive degree of freedom. In flat ΛCDM, μ, χ, R, J

and ξ are all functions of $\Omega_m(z)$ alone, so any apparent value added by a higher-order coordinate over a lower-order one signals leakage — residual dependence re-entering through a nonlinear reparameterization — rather than new content. Genuine content requires leaving the regime in which the reduction holds; for the cosmological case that means $w_{\rm DE} \neq -1$, where the rapidity curvature $d^2\xi/(d\ln s)^2 = -\frac{3}{2}w'_{\rm DE}$ is nonzero [6]. This is a screen to be applied before a test is run, not an explanation offered after one fails.

Safeguard 8.7 (Order and level). Conditions relating an amplitude-level quantity such as h to a ratio-level quantity such as a/b (Remark 2.6) must be labeled as partition-intrinsic relational conditions rather than physical balance laws.

9 Epistemic Tiers

Tier 1 — exact mathematics.

- the constraint $a + b = 1$ with $a, b > 0$;
- the altitude $h = \sqrt{ab}$ and the identity $4h^2 + \chi^2 = 1$;
- the response and rapidity coordinates and all coordinate transformations among the coordinates (a, b, h, χ, R, ξ);
- the intrinsic crossing $h = a/b$, its equivalence to $a = b^3$, its uniqueness, and the sign law of Lemma 2.4;
- the resulting cubic relations $b^3 + b - 1 = 0$ and $h^3 + 2h^2 + h - 1 = 0$, the supergolden coordinates $(a_*, h_*, b_*) = (\psi^{-3}, \psi^{-2}, \psi^{-1})$, and the algebraic identity of the compactness and altitude polynomials;
- the algebraic consequence (14) of imposing $a = b^3$ under the embedding $a/b = C$;
- the Fisher metric (18) and the numerator/denominator split (19).

Tier 2 — domain diagnostic.

- an independently motivated identification of the two physical sectors, including the reason they exhaust a total;
- a derived map $a/b = F(X)$, with its conventional content declared;
- the physical location X_* of a geometric landmark, reported together with its sensitivity to the normalization of F;
- a falsifiable test: a held-out prediction, a family collapse with a reported residual, an independently determined structure that then predicts a trajectory, or cross-domain recurrence under pre-declared normalization.

Nothing in the perturbative-GR example reaches Tier 2 in this note: the embedding (12) is schematic (Observation 5.1) and no test is performed here.

Tier 3 — speculative interpretation.

- claims of a universal mechanism behind the recurrence of the partition;
- claims that the intrinsic cubic is fundamental, or that its root is a constant of nature;
- claims that distinct physical domains share dynamics because they share partition geometry;
- claims that a landmark marks a sharp physical onset.

10 Discussion

Three points deserve emphasis.

First, the architecture explains why the programme's strongest and weakest results look superficially alike. Both consist of a partition, an embedding, and a landmark. They differ in exactly one place: whether the embedding was derived or chosen. The perturbative-GR case is the strongest available because the decomposition (9) is exact and the ratio (11) pre-exists the partition; it remains short of a diagnostic because (11) is schematic. The two-body case is the weakest not because its embedding is doubtful — it is exact — but because the domain assigns the landmark no role.

Second, the architecture makes the failure modes explicit and separable. If a partition is not genuine, step 1 fails and nothing downstream can be repaired. If the embedding is chosen rather than derived, step 5 fails and any numerical agreement is manufactured. If the landmark is real and the embedding is derived but the domain distinguishes nothing there, the result is a transit marker: a correct, exactly located, physically empty statement. These are three different negative results, and distinguishing them is more useful than aggregating them into a single verdict.

Third, the cases now on the table separate cleanly along these axes.

Case	Embedding	Domain role for landmark	Occupancy
Perturbative GR, C_{crit} (§4)	schematic: $a/b \sim C$ up to an unpinned normalization (Obs. 5.1)	not established	untested here
ΛCDM, R_J (§7.1(a))	derived from the conservation laws	yes: extremum of the flow acceleration	no ($z \approx 1.0$, -0.17)
ΛCDM, cubic (§7.1(b))	derived from the conservation laws	provably none: linear $U(\chi)$	yes ($z \approx 0.002$)
Two-body, cubic (§7)	exact: $F(q) = q$	none	not applicable

Fourth, a comparison drawn in the earlier cosmological work should be read the other way round. That work contrasted the linear cosmological potential (26) with a compactness flow in which the cubic is an attracting state, which invites the reading that ΛCDM is the weaker case. Under the present architecture it is the stronger one. The cosmological embedding is derived — (23) follows from the conservation laws alone — and it returns a clean, provable negative. The compactness flow's potential, by contrast, is a normal form obtained by quadrature and carries no information beyond the sign field of the flow; its critical point is inherited from the partition geometry rather than derived from general relativity, and the current form of that result asserts robustness across a class of positive prefactors rather than selection [7]. The two cases therefore divide the labor differently: ΛCDM has the better embedding and an empty landmark, while perturbative GR has the more suggestive landmark and a schematic embedding with an unpinned normalization. Neither is a diagnostic, and the useful statement is the pair rather than either one alone.

Fifth, negative results of the third kind are informative in the same way that a failed family test is informative in the dynamical setting [2]. A domain in which every intrinsic landmark is a transit marker is a domain whose physics is not organized by its partition structure; that is a fact about the domain, stated in a common language, and it is worth recording.

What the framework offers, then, is not derivation but organization with declared seams: a fixed constrained state space, a coordinate vocabulary that translates between bounded, hyperbolic and response descriptions, a short list of exactly located intrinsic landmarks, and a protocol that makes the physical content — and the burden of proof — reside entirely in the embedding.

11 Conclusion

A conserved two-sector system may admit a universal partition geometry, but that geometry does not determine the system's dynamics. It supplies invariant coordinates, coupling measures, asymmetry measures, and intrinsic relational landmarks. Domain physics supplies the sectors, the embedding, the evolution, and the empirical meaning. A physical diagnostic emerges only where these two layers intersect under an independently justified map.

This is also why the original construction required both ingredients at once, and why neither could have been dispensed with. The Einstein equations grounded the partition in a physically meaningful linear–nonlinear decomposition, while the Thales semicircle supplied the intrinsic relational condition that selected the cubic landmark. Without the field equations there would have been no reason to regard the two sectors as a genuine partition of anything; without the semicircle there would have been no condition to impose on it, and the operator scaling alone yields only the trivial bound $C < 1$. The cubic diagnostic is the intersection, and it is no firmer than the weaker of the two ingredients — at present, the schematic normalization of the sector ratio.

12 Relation to Prior Work

This note synthesizes and clarifies results already developed in earlier work; it introduces no new physical law, no new theorem, and no new empirical claim. Specifically:

- The coordinate vocabulary and the multi-face translation between bounded, hyperbolic and response descriptions were developed in [1].
- The Fisher kinetic skeleton, the universal-denominator / contextual-numerator split, the normal-form status of single-trajectory extraction, and the over-determination routes cited in §6 and §8 were developed in [2].
- The altitude–ratio crossing, its equivalence to $a = b^3$, the uniqueness of the interior solution, and the supergolden coordinates of Proposition 2.3 were established in [3].
- The compactness embedding $a/b = C$, the operator rearrangement (9), and the cubic (14) were developed in [4]; the transversality of the crossing and the prefactor-independence of the associated relaxation class, together with the recalibration of language from "dynamically selected" to "attracting equilibrium of a modelling class," also appear there.
- The similarity-composition reading of the structural content of the field equations, and the explicit statement that the field equations are not derived from the geometry, appear in [5].
- The exact flat-ΛCDM flow identities (24), the landmark $R_J = 3/2$, the rapidity identity (23), the Fisher-gradient representation (26) with its linear potential, and the numerical redshift placement of all cosmological landmarks were established in [6]. The present note adds only the reading of that paper's two landmarks as opposite failure modes (Interpretation 7.2), and the ordering correction of Remark 7.1.
- The numerical crowding of the region near 0.68 was recorded in [9]; Safeguard 8.5 generalizes it beyond cosmology.
- **Citation hygiene.** The June 2026 note [8] is superseded by [7], which withdraws the $C(1-C)$ prefactor and replaces "dynamically selected" with "attracting equilibrium of the imbalance-driven class." Forward citations should point to the revised note, so that the withdrawn artifacts are not carried into later work.

- **Second hygiene item.** The Blanchet living review has been superseded by its current edition [14]; earlier notes in this programme cite the 2014 edition, and forward citations should be repointed. More importantly, one of those notes attributes to that review a numerical statement — that post-Newtonian accuracy degrades for compactness above roughly 0.4–0.5 — which the review does not supply; the source supports the qualitative loss of accuracy approaching merger and no threshold value. That attribution should be weakened to the qualitative form or removed.

- The mass-partition organization of Newtonian two-body quantities, the altitude theorem $h^2 = m_1 m_2$, and the coupling–timing asymmetry were developed in [10].

The contribution of the present note is organizational: it names the two layers, states the division of labor (1), separates the intrinsic marker from its embedding and its interpretation (§5), and records the safeguard that an arbitrary embedding can manufacture any threshold (Safeguard 5.2). Where earlier work stated these points locally, in the context of a particular calculation, they are here stated once, generally, and in a form that applies before the next calculation is begun.

Acknowledgment

The author acknowledges the use of artificial-intelligence tools for editorial assistance, structural organization, algebraic checking, and consistency review. All conceptual development, research direction, interpretation, and final responsibility for the content belong to the author.

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